

VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

LAB PROJECT REPORT



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CERTIFICATE

This is to certify that the Object Oriented Programming through Java Lab project report entitled “ **Electricity Bill Management System**” submitted by **Sunhera Fathima (24911A1258)** to Vidya Jyothi Institute of Technology (An Autonomous Institution), Hyderabad, of B. Tech. II year, II Semester in Information Technology a Bonafide record of Lab project work carried out by them under my supervision.

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Abstract

The Electricity Bill Management System is a Java-based application developed using JDBC and MySQL for efficient management of customer electricity billing records. In traditional systems, electricity bill calculations and customer data management are often performed manually, which can lead to errors, increased workload, and time consumption. Manual processes may also create difficulties in maintaining records and retrieving customer information when required.

The main objective of this project is to automate electricity bill generation and customer record management. The system accepts customer details such as Customer ID, Customer Name, Meter Number, and Units Consumed. Based on the number of units consumed, the application automatically calculates the electricity bill amount and stores all customer details in a database for future use.

The technologies used in this project include Java programming language, JDBC (Java Database Connectivity), MySQL database, MySQL Workbench, and Visual Studio Code. Object-Oriented Programming concepts such as classes, objects, and exception handling are also implemented.

The expected outcome of this project is to provide an efficient and reliable system that reduces manual effort, improves calculation accuracy, minimizes errors, and ensures secure storage of customer billing information. The system enables easy insertion, retrieval, and management of customer records while improving overall efficiency.

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Chapter 1 – Introduction

1. Project Overview

The Electricity Bill Management System is a Java-based application developed to automate the process of electricity bill calculation and customer record management. The system allows users to enter customer details such as Customer ID, Customer Name, Meter Number, and Units Consumed. Based on the number of units consumed, the application automatically calculates the bill amount and stores the details in a MySQL database using JDBC connectivity.

The project reduces the need for manual calculations and record maintenance. Manual billing systems may lead to errors, data loss, and increased processing time. This system provides a reliable and efficient solution for maintaining customer billing records.

Real-world

Application:

The system can be used in electricity departments and organizations for managing customer information and generating electricity bills accurately. It helps reduce paperwork, saves time, and improves the efficiency of billing operations.

2. Objectives

The main objectives of the Electricity Bill Management System are:

- To automate manual electricity billing operations
- To reduce calculation errors
- To improve efficiency in managing customer records
- To provide fast and accurate bill generation
- To store customer information securely in a database
- To reduce paperwork and manual effort

1.3 Scope of the Project

The scope of this project includes managing customer billing information and automating electricity bill calculations. The system can be used by electricity service providers and administrators for storing and retrieving customer records efficiently.

Future scalability of the system includes adding advanced features such as graphical user interfaces, online payment facilities, customer authentication systems, report generation, and mobile application support. These enhancements can improve usability and expand the functionality of the project.

Chapter 2 – System Analysis

1. Functional Requirements

The Electricity Bill Management System performs the following functions:

- Add customer records
- Store customer details in database
- Calculate electricity bill automatically
- Retrieve customer information
- Connect Java application with MySQL using JDBC

2.2 Non-Functional Requirements

The non-functional requirements describe the quality and performance of the system.

- Security – Customer data should be stored securely in the database
- Performance – The system should calculate and store records quickly
- Reliability – The application should provide accurate results without failures
- Scalability – The system should support future enhancements and additional features
- Usability – The application should be simple and user-friendly

2.3 Software Requirements

Requirement Details

OS Windows/Linux

Language Java

IDE VS Code

Database MySQL

Connectivity JDBC

JDK JDK 21

2.4 Hardware Requirements

Requirement Details

RAM 4 GB

Requirement Details

Processor Intel i3 or above

Storage 500 GB

Chapter 3 – System Design

1. Architecture Diagram

The architecture of the Electricity Bill Management System consists of four main components:

User → Java Application → JDBC → MySQL Database

- User enters customer details
- Java application processes the input
- JDBC connects the application with MySQL database
- Database stores customer and bill information

3.2 UML Diagrams

The following UML diagrams are used in this project:

- Use Case Diagram – Shows interaction between user and system
- Class Diagram – Represents classes and their relationships
- Sequence Diagram – Shows the sequence of operations in the system
- Activity Diagram – Represents the workflow of bill generation

3.3 Database Design

The project uses one main table to store customer and billing information

Table Name	Purpose
------------	---------

customers	Store customer details and bill amount
-----------	--

Fields in customers table:

- customerId, customerName, meterNumber, unitsConsumed, billAmount

Chapter 4 – Implementation

4.1 Modules	Description
Customer Module	Stores customer details
Bill Module	Calculates electricity bill amount
Database Module	Connects Java application with MySQL
Main Module	Accepts user input and executes operations

4.2 Important Java Concepts Used

- Classes & Objects
- Exception Handling
- File Handling
- JDBC Connectivity
- Methods and Constructors
- Conditional Statements

4.3 Code Snippets

Database Connection Code:

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

```
Connection con = DriverManager.getConnection(  
"jdbc:mysql://localhost:3306/electricitydb",  
"root",  
"" );
```

Bill Calculation Code:

```
if(units <= 100)  
{  
    bill = units * 2;  
}  
  
else if(units <= 300)  
{  
    bill = (100 * 2) + ((units - 100) * 3);  
}  
  
else  
{  
    bill = (100 * 2) + (200 * 3)  
        + ((units - 300) * 5);} 
```

Chapter 5 – Testing

5.1 Test Cases

Test Case ID	Input	Expected Output	Result
TC01	Valid customer details	Customer added successfully	Pass
TC02	Invalid database connection	Error message displayed	Pass
TC03	Valid units consumed	Correct bill amount generated	Pass
TC04	Invalid input values	Exception handled	Pass

5.2 Error Handling

The following exceptions are handled in the project:

- SQLException
- NumberFormatException
- FileNotFoundException
- InputMismatchException

Chapter 6 – Results and Screenshots

```

src > electricitybill > J Main.java > % Main > main(String[])
5  public class Main {
7      public static void main(String args[]) {
9          Scanner sc = new Scanner(System.in);
10
11         System.out.println(x: "Enter Customer ID:");
12         int id = sc.nextInt();
13
14         sc.nextLine();
15
16         System.out.println(x: "Enter Customer Name:");
17         String name = sc.nextLine();
18
19         System.out.println(x: "Enter Meter Number:");
20         String meter = sc.nextLine();
21
22         System.out.println(x: "Enter Units Consumed:");
23         int units = sc.nextInt();
24
25         Customer c = new Customer(id, name, meter, units);
26     }
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\sirim\OneDrive\Desktop\ElectricityBillManagement\ElectricityBillManagement> & 'C:\Users\sirim\AppData\Local\Temp\cp_3r16wmwpt22zly52my3col'
'es\Java\jdk-21.0.11\bin\java.exe' '@C:\Users\sirim\AppData\Local\Temp\cp_3r16wmwpt22zly52my3col' 'electricitybill.Main'
Enter Customer ID:
3
Enter Customer Name:
Sindhu
Enter Meter Number:
M678
Enter Units Consumed:
450
Customer added successfully

```

Figure 6.1: Execution output screen displaying customer input and successful bill generation

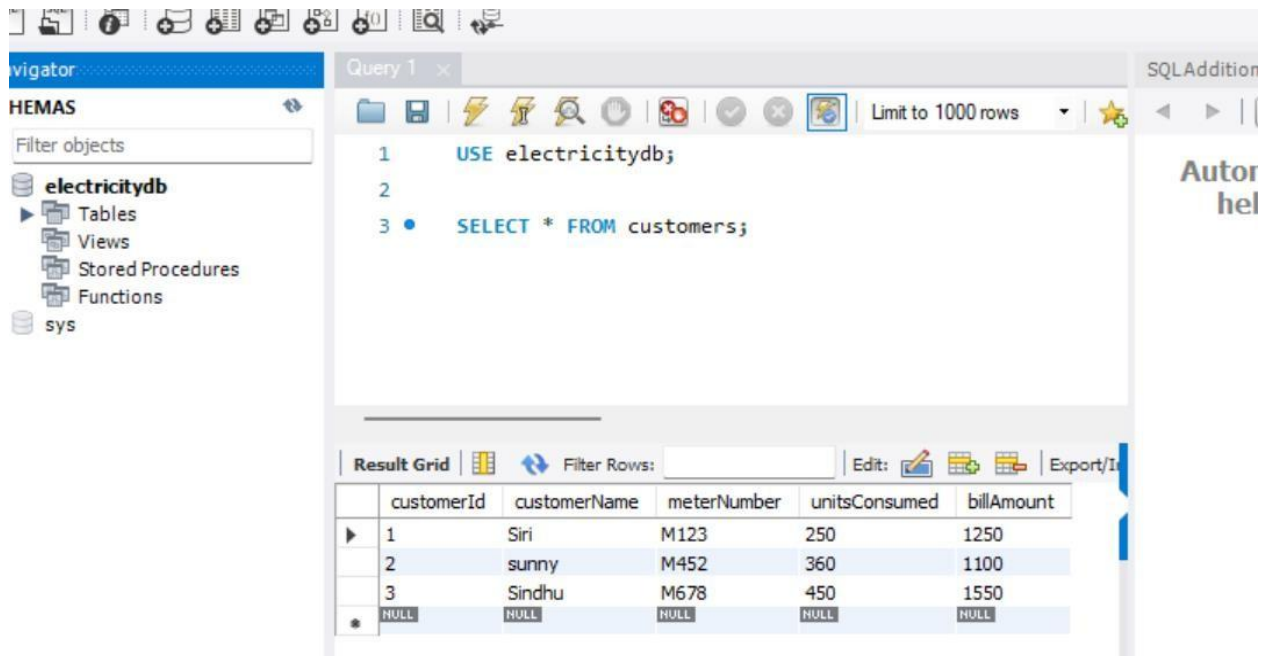


Figure 6.2: Database creation and table structure in MySQL Workbench

siri-mylarishetty Resolved merge conflict

d729a30 · 19 hours ago
3 Commits

.vscode	Electricity Bill Management using Java JDBC	19 hours ago
bin/electricitybill	Electricity Bill Management using Java JDBC	19 hours ago
lib	Electricity Bill Management using Java JDBC	19 hours ago
src/electricitybill	Electricity Bill Management using Java JDBC	19 hours ago
README.md	Resolved merge conflict	19 hours ago

README

ElectricityBillManagement

Electricity Bill Management System using Java, JDBC, and MySQL with automatic bill calculation and customer managementgit add README.md

Electricity Bill Management System using Java, JDBC, and MySQL with automatic bill calculation and customer management.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published
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Packages

No packages published
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Figure 6.3: GitHub repository containing the Electricity Bill Management project files

Chapter 7 – Conclusion and Future Scope

Conclusion

The Electricity Bill Management System was successfully developed using Java, JDBC, and MySQL. The project helps automate the process of storing customer details and calculating electricity bills efficiently. Manual calculations and record maintenance are reduced, which minimizes errors and saves time.

Through this project, important Java concepts such as Classes & Objects, JDBC connectivity, Exception Handling, and Database Management were implemented practically. The project also improved understanding of SQL queries, GitHub integration, and real-time database operations.

Overall, the system works effectively for managing customer electricity billing records in a simple and user-friendly manner.

Future Enhancements

The project can be further improved with additional features such as:

- Developing a graphical user interface using Java Swing or JavaFX
- Adding user login and authentication system
- Integrating online payment gateway support
- Creating a mobile or web-based version of the application
- Generating PDF bill reports automatically
- Adding cloud database integration for remote access
- Implementing search, update, and delete functionalities for customer records

References

1. **Java: The Complete Reference**
Herbert Schildt, Oracle Press.
2. **MySQL Workbench**
3. [Oracle Java Documentation](#)
4. [MySQL Official Documentation](#)
5. [GitHub Official Website](#)

